

ZICHEN LIU

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Computer Science Ph.D. student at Arizona State University researching software security (program analysis and automated vulnerability repair), social-engineering security, and the security of LLM-based systems.

EDUCATION	Arizona State University	Tempe, AZ
	<i>Ph.D. in Computer Science, advisor Prof. Xusheng Xiao</i>	2023–May 2028 (expected)
	University of Washington	Seattle, WA
	<i>Master of Science in Electrical & Computer Engineering</i>	2019–2021
	Cuiying Honors College, Lanzhou University	Lanzhou, China
	<i>Bachelor of Science in Physics, with highest Honors</i>	2015–2019

- PUBLICATIONS
1. **Zichen Liu** and Xusheng Xiao. “APPBDS: LLM-Powered Description Synthesis for Sensitive Behaviors in Mobile Apps.” *Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2025. [Paper]
 2. Shang Ma, Chen Yanai, Avichai Ben, **Zichen Liu**, Yanfang Ye, Xusheng Xiao. “Demystifying Scam Playbooks: Psychological Techniques, Scenario Taxonomy, and Conversation-Aware Detection.” *Under review at the IEEE Symposium on Security and Privacy (S&P)*, 2027. [arXiv]
 3. **Zichen Liu**, Shao Yang, Xusheng Xiao. “BINCTX: Multi-Modal Representation Learning for Robust Android App Behavior Detection.” *arXiv preprint arXiv:2510.14344*, 2025. [arXiv]
 4. Jiongxiao Wang*, **Zichen Liu***, Keun Hee Park, Zhuojun Jiang, Zhaoheng Zheng, Zhuofeng Wu, Muhao Chen, Chaowei Xiao. “Adversarial Demonstration Attacks on Large Language Models.” *arXiv preprint arXiv:2305.14950*, 2023. (*equal contribution) [arXiv]
 5. Cheng-Yen Yang*, Aotian Zheng*, **Zichen Liu**, Yujie Li, Jenq-Neng Hwang. “Tracklet-based Multi-camera Multiple People Tracking.” *Challenge entry (winner talk), ICCV 2021 Multi-camera Multiple People Tracking Workshop*. [Presentation]

- RESEARCH EXPERIENCE
- Research Assistant | RISE Lab @ASU (Prof. Xusheng Xiao)** 2023.7–present
- Android Program Analysis & Security**
- APPBDS (ASE 2025): LLM-powered descriptions of how Android apps use sensitive permissions.
 - Static analysis builds an inter-component call graph fused with UI context (UI-Fused Call Graph); privacy-relevant node patterns shrink it from $\sim 46K$ nodes to under 100, and specialized LLM agents cross-check code evidence, privacy-policy propositions and similar-app behavior before synthesizing the description
 - On 270 real-world apps: +22.0% BERT-Score and $8.3\times$, $6.2\times$ and $7.6\times$ higher richness, specificity and semantic relatedness than the prior state of the art; +5.2%, 34.9%, 28.0% and 55.4% over a GPT-4o prompting baseline; robust across five open LLMs and to code obfuscation
 - BINCTX: multi-modal detection of malware and undesired app behaviors.
 - Fuses a bytecode-as-image view, manifest/URL context and third-party-library usage counted along call paths into one context-aware classifier
 - 94.73% macro-F1 on $\sim 17K$ apps, ≥ 14.9 points over the strongest baseline, 84% F1 after commercial obfuscation
- Social Engineering & Online Scam Defense**
- EARS, *Early Anti-scam Recognition System (IEEE S&P 2027, under review)*: identifies the scam scenario behind a customer’s narrative while the conversation is still unfolding.
 - A scenario classifier fine-tuned on conversations with turn-level supervision, a per-turn phase classifier that withholds judgment on benign turns and commits once diagnostic evidence appears, and label-preserving LLM data augmentation

- On 14.4K conversation turns: tactic/scenario macro-F1 of 84.1/71.4 over 6 tactics and 18 scenarios, +13.2/+25.6 over few-shot GPT-5.4; per-turn accuracy over $\sim 56\%$ at the first scam evidence. Tested in real bank customer-service operations; deployment has begun
- *Longitudinal measurement of the scam ecosystem* [Ongoing]: built a 611K-report scam-incident corpus spanning 2018–2026, and traces how scam scenarios, campaign organization, contact channels and persuasion effectiveness evolve over time and across regions, and respond to external events

Security Knowledge for LLM-Guided Vulnerability Repair [Ongoing]

- *Security impact $\times$ fix pattern knowledge base*: links the security constraint a vulnerability violates to the patch that fixes it.
 - Designed a mechanism-level security-impact taxonomy that captures the concrete safety rules violated during vulnerability execution; it synthesizes existing security-impact and bug taxonomies and consolidates them on empirical data into 12 base categories with finer-grained subtypes
 - Built a fix-pattern mining pipeline over 12K security patches: patches normalized to function-level pre/post versions, GumTree edit scripts turned into diff-ASTs, TreeCen centrality embeddings, and UMAP/HDBSCAN clustering into candidate fix-pattern families

Research Assistant | SaFo Lab @ASU (Prof. Chaowei Xiao)

2023.2–2023.7

LLM Security & Safety

- *advICL*: adversarial demonstration attacks on in-context learning.
 - A black-box attacker perturbs only the in-context demonstrations, leaving the query untouched, to induce targeted misclassification; Transferable-advICL iterates the attack over a candidate set so one adversarial demonstration transfers to unseen inputs
 - On GPT2-XL, LLaMA-7B and Vicuna-7B over SST-2, TREC, DBpedia and RTE: attack success up to 97.72%, rising with more demonstrations; 72.32% transfer success on unseen inputs; perturbed demonstrations kept high perceptual quality ($\sim 90\%$ human-rated)

Research Assistant | Information Processing Lab @UW (Prof. Jenq-Neng Hwang) 2020.12–2021.9

Multi-Camera Multiple People Tracking

- *Tracklet-based multi-camera tracking*: built a system whose cross-camera tracklet association extends a single-camera tracker (FairMOT) across views; 3rd place in the ICCV 2021 MMP-Tracking Challenge (winner talk)

INDUSTRY EXPERIENCE

Machine Learning Engineer | Ernst Leitz Labs, Leica Camera AG

2021.9–2022.11

- Developed deep-learning replacements for stages of the mobile camera ISP pipeline, and built the rendering engine and depth estimation behind the Leitz Looks Engine
- Ported depth models such as RAFT-Stereo and MiDaS onto a next-generation Qualcomm Snapdragon chipset, with speed and power optimization and performance profiling

TEACHING EXPERIENCE

Teaching Assistant | Arizona State University

- CSE 464 Software Quality Assurance and Testing Fall 2023–Fall 2025
- CSE 471 Introduction to Artificial Intelligence, Spring 24

ACADEMIC SERVICE

- **Artifact Evaluation Committee**: IEEE S&P 2027, ISSTA 2026, ASE 2026

- **Reviewer**: NeurIPS 2026 FLLMPT Workshop

SKILLS

Languages: Python, Java, C/C++

ML & LLM Tooling: PyTorch, Hugging Face Transformers, PEFT, vLLM

Program Analysis & Security: Soot/SootUp, FlowDroid, Android/APK Analysis, Call-Graph & Taint Analysis

Tools & DevOps: Git, Docker, CI/CD